

# Machine Learning in Materials Chemistry Trinity Term 2026

## Assignment 3: Comparison of empirical potential and MLIP

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### Introduction

Interatomic potentials are central to atomistic simulations because they determine how the energy and forces of a structure are calculated. Traditionally, empirical potentials such as the Tersoff potential [7] have been used for carbon systems, where their functional form is fitted to reproduce selected reference properties. More recently, machine-learning-based interatomic potentials (MLIP) have become an alternative approach, as they can learn more flexible relationships from reference data and may therefore describe complex bonding environments more accurately.

In this task, the behaviour of a MLIP is compared with that of an empirically fitted potential for carbon structures. The molecular dynamics (MD) simulations were carried out with ASE ([1, 3]). The comparison focuses on the stability of simulations carried out with either the ML or the empiric potential, including annealing, geometry optimisation, graphitisation and melt-quench simulations.

### Methods and Reproducibility

The empirical potential used in this project is the Tersoff potential ([7]), which is particularly useful to simulate diverse carbon structures as given in this task. ASE requires the parameters for this potential to be given when running a simulation. These parameters were mostly taken from [7], and complemented by comparing them to the corresponding ones used in LAMMPS ([4, 8]). Note that the difference in cutoff from the Tersoff potential (here parameterised by  $R$  and  $D$ , slightly differently to the original paper - see the documentation of LAMMPS for more detail) to the MLIP does not pose an issue as the cutoff as defined in the training of the MLIP does not strictly mean that atoms that are closer together than this cutoff are bound, but simply that they might have any kind of influence on one another.

**Table 1: Tersoff parameters used for C-C-C interactions.**

| Parameter   | Value                    |
|-------------|--------------------------|
| $m$         | 3.0                      |
| $\gamma$    | 1.0                      |
| $\lambda_3$ | 0.0                      |
| $A$         | $1.3936 \times 10^3$     |
| $B$         | $3.4674 \times 10^2$     |
| $\lambda_1$ | 3.4879                   |
| $\lambda_2$ | 2.2119                   |
| $\beta$     | $1.5724 \times 10^{-7}$  |
| $n$         | $7.2751 \times 10^{-1}$  |
| $c$         | $3.8049 \times 10^4$     |
| $d$         | 4.384                    |
| $h$         | $-5.7058 \times 10^{-1}$ |
| $R$         | 1.95                     |
| $D$         | 0.15                     |

The dataset used in this project is the C-GAP-20U dataset ([5]) provided in ASE and containing a variety of different carbon structures. For question 3a, three different sample structures representing different configurations and therefore displaying different behaviour were chosen for the MD simulations (Amorphous\_Bulk, Crystalline\_RSS and Graphite), making sure that they were not previously part of training dataset for the MLIP. For question 3b, five different sample structures from the same configuration type Amorphous\_Bulk (for comparability) were chosen to represent five different densities of  $\rho \in \{2.05, 2.3, 2.6, 2.9, 3.2\} \text{ g cm}^{-3}$

A few adjustments to the parameters from the class were made, both to ensure correctness and ensure computational tractability given the limited computational resources. These include that for each simulation, the original starting structure is used as an initial frame instead of overwriting this starting structure and therefore continuing with the end state from an earlier simulation.

The MLIP was trained using graph-pes ([2]): The bespoke-train model from the classes was used, the corresponding configuration file can be found in

train-bespoke.yaml. As explored in the class already, this model is unfortunately not completely converged. The root mean squared error of the potential energy for this MLIP is evaluated to be 176.01 meV / atom. A more thorough hyperparameter search and longer training was not possible due to computational limitations.

The versions of all programs used (for reproducibility of the results) can be found in the file environment.yaml. Additionally, the basic version of Ovito version 3.15.4 was used for visualisations ([6]).

### Question 3a)

#### Annealing

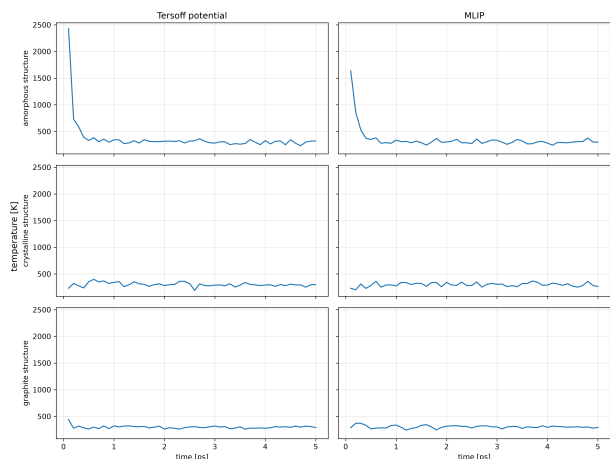
A first stability test for any potential is the process of annealing: holding the simulation at constant temperature using the NVT ensemble. In this case, 300 K was used as a temperature for 5 ps. As can be seen in Figure 1 and 2, both potentials show the expected behaviour of producing constant (with the exception of small oscillations) energy and temperature trajectories after a short calibration/ relaxation phase for all three structures.

### Geometry Optimisation

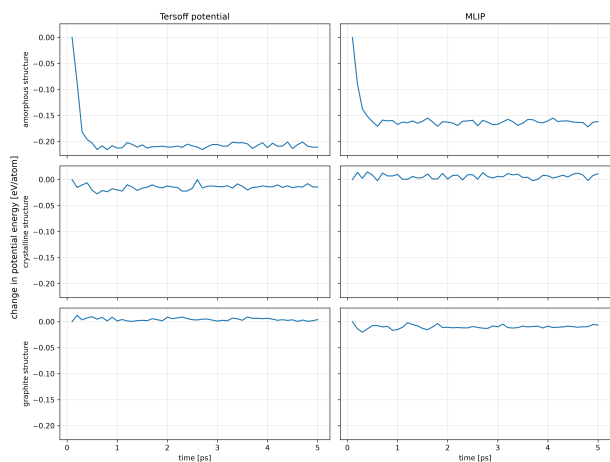
**Table 2: Energies per atom before and after geometry optimisation. All energies are given in eV/atom.**

| Structure   | $E_{\text{start}}/N$ | $E_{\text{end}}/N$ | $\Delta E/N$ |
|-------------|----------------------|--------------------|--------------|
| amorphous   | -5.5178              | -6.6871            | -1.1693      |
| crystalline | -6.7424              | -6.7964            | -0.0540      |
| graphite    | -7.3168              | -7.3907            | -0.0739      |

As expected from the larger difference in energy per atom before and after the geometry optimisation (see Table 2 and Table 3) and suspected because of the calibration phase seen in the annealing (see Figure 2), the amorphous structure changes visibly during the simulation (pay attention to the upper right corner in the visualisation, for example), in contrast to the graphite



**Figure 1: Temperature [K] over simulation time [ps] for three different structures and the two different potentials. All simulations show stable behaviour, staying at around the desired 300 K after a short calibration/ relaxation phase for the amorphous structure.**

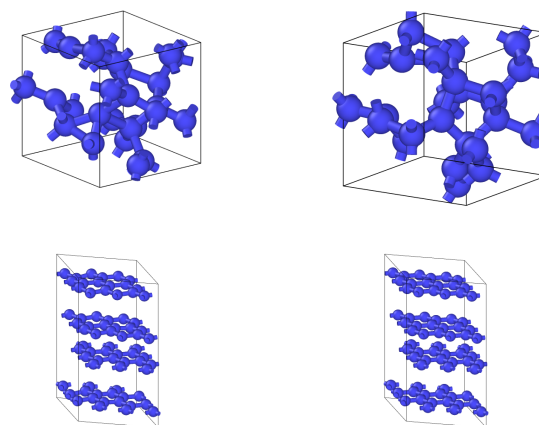


**Figure 2: Change in potential energy [eV/atom] (so potential energy relative to the potential energy at the initial frame, to eliminate offsets between structures) over simulation time [ps] for three different structures and the two different potentials. All simulations show stable behaviour, staying at a relatively constant energy after a short calibration/ relaxation phase for the amorphous structure.**

**Table 3: MLIP model: energies per atom before and after geometry optimisation. All energies are given in eV/atom.**

| Structure   | $E_{\text{start}}/N$ | $E_{\text{end}}/N$ | $\Delta E/N$ |
|-------------|----------------------|--------------------|--------------|
| amorphous   | -6.4468              | -7.2752            | -0.8284      |
| crystalline | -7.6078              | -7.6106            | -0.0028      |
| graphite    | -7.9407              | -8.0093            | -0.0685      |

structure which remains almost the same and apparently was already given in its optimal structure.



**Figure 3: Visualisation of the amorphous structure (top) and the graphite structure (bottom) before (left column) and after geometry optimisation (right column).**

## Graphitisation

Graphitisation was performed by imposing a temperature ramp at the start of the simulation from 300 K to 3000 K in 20 ps, and subsequently annealing at 3000 K for 200 ps.

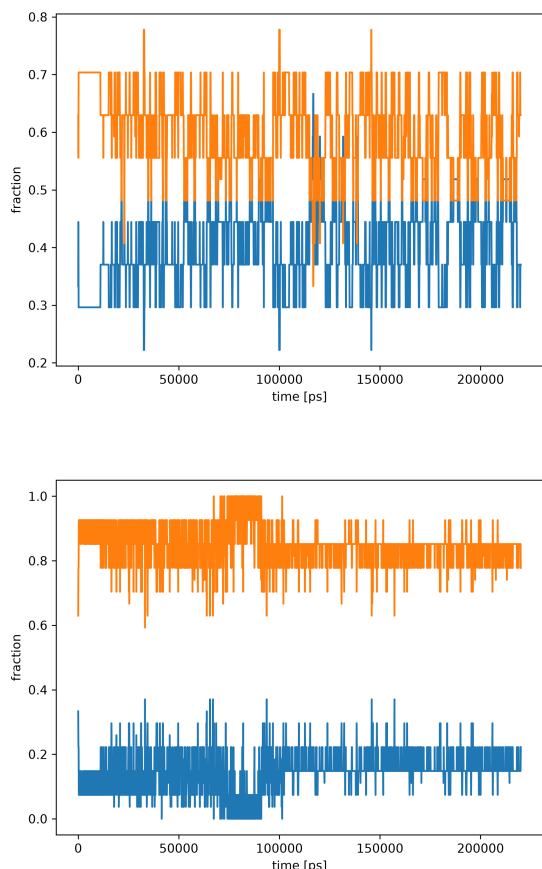
The coordination analysis in Figure 4 does not provide clear evidence for successful graphitisation. For the Tersoff potential, the  $sp^2$  and  $sp^3$  fractions fluctuate throughout the trajectory, but no sustained increase in the  $sp^2$  fraction is observed. Instead, both fractions remain within a similar range over time, suggesting that the system undergoes local bond rearrangements without forming a progressively more graphitic network.

The MLIP trajectory shows even less indication of graphitisation. The structure remains dominated by  $sp^3$ -coordinated atoms for most of the simulation, while the  $sp^2$  fraction stays comparatively low. This is inconsistent with the expected trend for graphitisation, where a gradual conversion towards threefold-coordinated,  $sp^2$ -rich carbon would be expected. Together with the visual inspection of the final structures, which does not show clear graphitic domains or layered ordering, this suggests that graphitisation was not achieved under the chosen simulation conditions.

The visualisation of the resulting structures also showed not visible graphitisation. Possible reasons include insufficient annealing temperature, limited simulation time, or "kinetic trapping" of the amorphous structure. It is also possible that the two potentials differ in how easily they allow bond breaking and reformation towards  $sp^2$ -rich structures.

## Question 3b)

Due to the Tersoff potential not being optimised in ASE (from the documentation: "The performance of the routines for this calculator have not been optimized for speed nor benchmarked with the LAMMPS implementation."), running long simulations for melt quenching posed a difficulty. Additionally, the Tersoff potential was not stable under extreme conditions of 4500 K or higher (division by zero error, the temperature fell to zero, possibly because of numerical instabilities). These limitations lead to the choice of the following simulation parameters, although the temperature might be a bit low for melting - something to keep in mind when interpreting the results: initial and end temperature

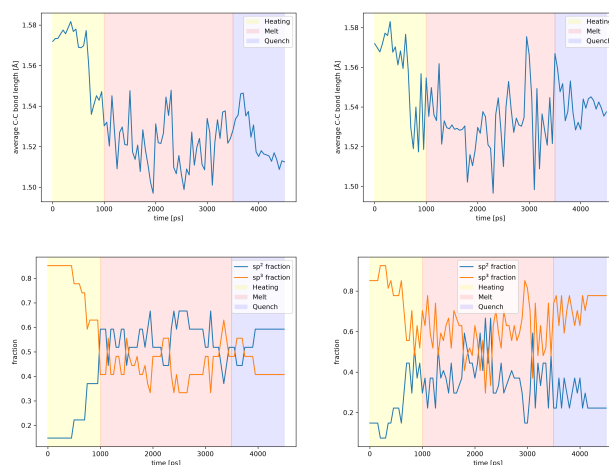


**Figure 4: Evolution of the  $sp^2$  and  $sp^3$  coordination fractions during the graphitisation annealing simulation of the amorphous carbon structure. The upper panel shows the results obtained with the Tersoff potential, while the lower panel shows the corresponding MLIP simulation. Although graphitisation would be expected to increase the  $sp^2$  fraction and reduce the  $sp^3$  fraction over time, both trajectories mainly show fluctuations around approximately stable coordination fractions.**

of 300K, melting temperature 4000 K, 2000 heating, 5000 melting and 2000 quenching steps, each with a timestep of 0.5 fs.

For the analysis of coordination numbers, a cutoff length of 1.9 was chosen.

Figure 5 shows the trajectory of average bond lengths and fractions of  $sp^2$ - and  $sp^3$ -coordinated atoms during melt-quench simulations of amorphous carbon at density  $\rho \approx 3.2 \text{ g cm}^{-3}$ . During heating, an increase in the average C–C bond length might be expected due to thermal expansion and enhanced atomic motion. However, the observed decrease suggests that the average is strongly affected by structural rearrangements and by the cutoff-based definition of bonds. During the melt stage, longer or weakly bonded neighbours may no longer be counted as bonded, while shorter  $sp^2$ -like bonds can become more prevalent. Therefore, the average bond length should be interpreted together with the coordination fractions rather than as a direct measure of volume expansion alone.



**Figure 5: Evolution of the average C–C bond length and the fractions of  $sp^2$ - and  $sp^3$ -coordinated atoms during melt-quench simulations of amorphous carbon at density  $\rho \approx 3.2 \text{ g cm}^{-3}$ . The shaded regions indicate the heating, melt and quench stages of the simulation. The average bond length fluctuates strongly during the high-temperature melt stage and stabilises during quenching, while the coordination fractions reflect the restructuring of the carbon network over the course of the simulation.**

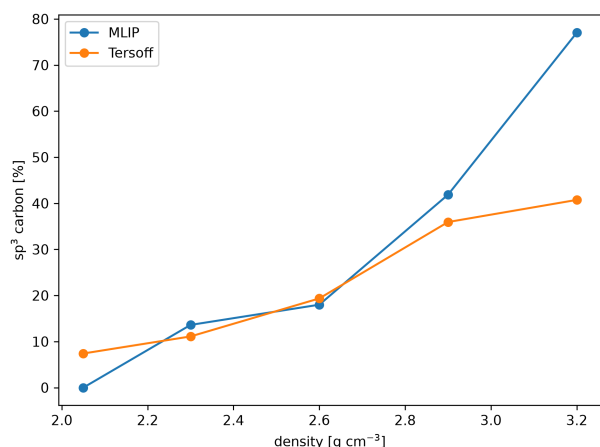
The coordination fractions support this interpretation. During the heating and melt stages, the fractions of  $sp^2$ - and  $sp^3$ -coordinated atoms fluctuate strongly, indicating continuous bond breaking and reformation. A decrease in the  $sp^3$  fraction together with an increase in the  $sp^2$  fraction suggests that the initially more tetrahedral network becomes less diamond-like and more graphitic or liquid-like during the melt. During the quench, these fluctuations become smaller and the coordination fractions gradually stabilise again, meaning that the local bonding environments are progressively frozen into the final amorphous structure.

Comparing the trajectories produced by the MLIP with the Tersoff potential shows that they both achieve very similar results in this simulation. One difference is that the fractions of  $sp^2$ - and  $sp^3$ -coordinated atoms fluctuate a bit more with the MLIP potential which might be because the potential is not stable enough yet (further training might minimise these issues). A comparison of various accuracy metrics for the MLIP and Tersoff potential could also be a starting point for such a further analysis.

The final  $sp^3$  fraction increases with density for both potentials, which is consistent with the expected behaviour of amorphous carbon. At lower densities, the structures are more open and contain mainly  $sp^2$ -like bonding parts, whereas higher densities favour more compact, tetrahedral  $sp^3$ -like coordination.

Both models show a similar trend up to intermediate densities. However, at higher densities the MLIP predicts a much stronger increase in the  $sp^3$  fraction than the Tersoff potential. In particular, at the highest density, the MLIP structure becomes strongly  $sp^3$ -rich, while the Tersoff model still predicts a substantially lower fraction of fourfold-coordinated atoms. This suggests that the two potentials differ in how they describe the relative stability of  $sp^2$  and  $sp^3$  bonding environments under compression.

The smoother and more moderate increase observed for the Tersoff model may indicate that the empirical potential favours less tetrahedral bonding in the investigated density range, or that it does not capture the density-driven transition towards



**Figure 6: Percentage of  $sp^3$ -coordinated carbon atoms in the final amorphous carbon structures obtained from melt-quench simulations at different starting densities (structures with different densities). The values were averaged over the final frames of each trajectory. Results are shown for both the MLIP and Tersoff potential.**

diamond-like amorphous carbon as strongly as the MLIP. This difference might also be influenced by the relatively large cutoff radius of cutoff = 3.7 used in the MLIP, which allows the model to include information from a larger local atomic environment and may therefore affect the stability of dense, highly coordinated structures. But, since the values were obtained from the end of finite melt-quench trajectories, they may also be affected by incomplete equilibration, limited simulation time, or insufficient melting before the quench.

## Conclusion and Outlook

Overall, the comparison between the empirical Tersoff potential and the MLIP shows that both approaches are able to produce physically meaningful qualitative trends for carbon structures, but also exhibit clear limitations under the investigated simulation conditions. In the annealing and geometry optimisation tests, both potentials produced stable trajectories and reasonable relaxation behaviour. The amorphous structure showed the largest energy decrease upon optimisation, while the graphite and crystalline structures were already closer to local minima.

The graphitisation test did not provide clear evidence for successful graphitisation. Instead of a sustained increase in the  $sp^2$  fraction, both models mainly showed fluctuations around approximately stable coordination fractions. This suggests that the chosen annealing temperature and simulation time were insufficient to form extended graphitic domains, or that the amorphous structures remained kinetically trapped.

In the melt-quench simulations, both potentials reproduced the expected qualitative trend that the final  $sp^3$  fraction increases with density. However, the MLIP predicted a stronger increase in  $sp^3$ -coordinated carbon at high densities than the Tersoff potential, indicating differences in how the two models describe dense, tetrahedral carbon environments. Since the melt temperature and simulation time were limited, these results should be interpreted as qualitative rather than fully equilibrated predictions.

It would be interesting to rerun these simulations with more computational resources and therefore longer simulation times

or a more fine-tuned MLIP - for example by fine tuning a larger model, investigating more exactly which structures exhibit the largest errors or running hyperparameter searches. Nevertheless, the results demonstrate how MLIPs and empirical potentials can both achieve physically meaningful results for carbon structures, but can differ not only in energy predictions, but also in the structural transformations they favour during atomistic simulations.

## References

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### **Declaration of AI usage**

For the coding elements in this assignment, artificial intelligence was used as assistance to learn about the tools used (in addition to their documentation, of course), as a starting point for the plots in question 3b) and in debugging, including the generation

of a few lines of code at a time. These were in any case always reviewed, understood and, if needed, thoroughly tested before incorporating them into the project. The structure and a vast majority of the code are created and written by myself without the assistance of AI and the work is clearly and substantively my own.